Jorge Ramón Parra Michel, Ph.D.

Mechanical Engineer | Optical Metrology Researcher

CONTACT

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NATIONALITIES

- Mexican | French

LANGUAGES

- Spanish — Native
- French — Native
- English — TOEFL 550

PROFILES

- 0000-0002-9145-5574
- Google Scholar
- Web of Science

PROFESSIONAL LICENCES

Ph.D. Licence No. 00019609
M.Sc. Licence No. 00019608
Mech. Eng. Licence No. 5384881

ENIC–NARIC EU Attestation

SOFTWARE & TOOLS

- MATLAB
- Python
- LaTeX / XeLaTeX
- SolidWorks
- ANSYS Workbench
- LabVIEW
- OpenCV
- AutoCAD / SpaceClaim
- MeshLab

Objective

To advance optical metrology and mechanical stress analysis through physics-based optical techniques and computational modeling. Expert in Electronic Speckle Pattern Interferometry (ESPI), Digital Image Correlation (DIC), and Structured Light Projection for 3D shape and deformation measurement. Skilled in Finite Element Analysis, precision manufacturing, and reverse engineering.

Professional Goals

To lead research in optical instrumentation, physics-informed deep learning for 3D topography reconstruction, and industrial metrology combining optics, mechanics, and AI. Strengthen the interface between scientific research, technological innovation, and higher education in engineering.

Fellowships & Recognition

2023–present — **SNI Level 1 — SECIHTI**
Mexican National System of Researchers

2025 — **Beca Santander–FIMPES Investigación**
Banco Santander Universidades & FIMPES México
Project: *Development and Assessment of Competencies in Advanced Manufacturing: AI Applications in Optical Inspection, Reverse Engineering, and Computational Analysis.*

Education

2007–2011 — **Ph.D. in Optics**
Centro de Investigaciones en Óptica (CIO), México
Thesis: *In-Plane ESPI for mechanical stress analysis and topometry using divergent illumination.* ENIC–NARIC: **Doctorat** (Level 8 EQF).

2004–2006 — **M.Sc. in Optics**
Centro de Investigaciones en Óptica (CIO), México
Thesis: *ESPI and phase shifting in dynamic events for analysis of uniaxial deformation in welds.* ENIC–NARIC: **Master** (Level 7 EQF, Ref. 2080168).

2000–2004 — **Mechanical–Electrical Engineer**
Universidad de Guadalajara, México
ENIC–NARIC: **Licence/Bachelor** (Level 6 EQF, Ref. 821055).

RESEARCH LINES

- ESPI / Interferometry
- Structured Light 3D
- Digital Image Correlation
- Physics-Informed DL
- Reverse Engineering
- Mechanical Design & FEA

PATENTS

- Wind Generator for Forced Draft Pipes (MX/2015/001586A)
- Machine to Separate Solid & Light Particles from Air (MX/2016/000865A)

TEACHING (CURRENT)

LEM430 — Steel Structures
MDM410 — 3D Simulation & Prototyping
LIB427 — Microcontrollers

Research Methodology

Professional Experience

2007–present

Professor–Researcher

Universidad La Salle Bajío, A.C., León, México

- Research in optical metrology, 3D shape measurement, stress–strain analysis.
- Design of interferometric and structured-light instrumentation.
- Computational tools (MATLAB, Python) for optical data processing.
- Teaching at undergraduate and graduate levels.
- R&D grant proposals and industry collaboration.

2019–present

Independent Consultant — 3D Optical Measurement

3D scanning and CAD modeling for mechanical, artistic, and industrial components using laser and structured-light systems for additive manufacturing and CNC production.

1998–2004

Mechanical Engineer (Jr.)

Industria Metálica de León, C.V.

Design, drafting, and structural analysis of mechanical assemblies.

Selected Publications

2026 **Parra-Michel, J.R.**, Martínez Santamaría, N.A., Aguilar Vargas, E.
Solving the Phase Unwrapping Problem in Dual Divergent Projection Systems Using a Physics-Informed Neural Network.
Proc. ICECET 2026, Rome, Italy (Paper ID 1199, accepted).

2026 **Parra-Michel, J.R.**, Martínez Santamaría, N.A., Aguilar Vargas, E.
Physics-informed hybrid U-Net for phase-to-height reconstruction in dual divergent structured-light systems (PIHUNN).
Optics and Lasers in Engineering, 201, 109693. doi:10.1016/j.optlaseng.2026.109693

2022 Hernández, A.R.C., Ontiveros, R.I.R., Ponce, H.E.E., Ramírez, J.M.O., **Parra-Michel, J.R.**, Moller, J.A.D.
Effect of silver on structural, optical, and electrical properties of ZnO:Al/Ag/ZnO:Al thin films.
Nova Scientia, 14(29). doi:10.21640/ns.v14i29.3038

2021 **Parra-Michel, J.R.**, Martínez-Peláez, R., Duarte-Moller, J.A.
Double structured light with divergent projection for surface topometry.
Meas. Sci. Technol., 32(8), 085203. doi:10.1088/1361-6501/abff80

2021 Montoya, M., Montelongo, Y., Jiang, N., Morris, S.M., **Parra-Michel, J.R.**
Free-form laser profilometry for pipeline inspection and 3D cylindrical reconstructions.
IEEE Sensors J., 21(24), 27855–27863. doi:10.1109/JSEN.2021.3130224

2020 **Parra-Michel, J.R.**, Gutiérrez-Hernández, D., Martínez-Peláez, R., Escobar, M.A.
A radial in-plane sensitivity interferometer with divergent illumination for displacement measurement.
Applied Sciences, 10(3), 908. doi:10.3390/app10030908

2020 Igba, V.M., Ahemen, I., et al., **Parra-Michel, J.R.**
Structural elucidation and optical properties of $LiZrO_2$–$LiBaZrO_3$ nanocomposite doped with Mn^{2+} ions.
Opt. Mater. Express, 10(11), 2877–2895. doi:10.1364/OME.402111